

Sheaves, Cosheaves and Applications

Summary

December 23, 2025

1 Problem Setup

This PhD thesis develops a comprehensive mathematical framework for sheaves and cosheaves as tools for organizing and analyzing data that varies over topological spaces. The author recognizes that while sheaves have been widely used in mathematics, their dual concept of cosheaves has been underexplored despite having important applications in data analysis, sensor networks, and persistent homology.

2 Thesis

The central thesis is that sheaves and cosheaves provide complementary perspectives on data organization: sheaves capture how local data can be restricted to smaller regions, while cosheaves capture how local data can be extended (glued) to larger regions. The thesis demonstrates that this duality is fundamental for understanding many modern data analysis problems, particularly those involving multi-scale analysis and network flows.

3 Key Concepts

Sheaves: Mathematical objects that assign data to open sets of a topological space with restriction maps that ensure consistency. Think of them as systems of local measurements that agree on overlaps.

Cosheaves: The dual concept to sheaves, assigning data to open sets with extension (gluing) maps. They capture how local information can be combined to form global understanding.

Cellular Sheaves/Cosheaves: Discrete versions defined on cell complexes, making them computable for practical applications.

Persistent Homology: A method for studying the shape of data across multiple scales, naturally expressed using sheaf and cosheaf theory.

Network Flows: Applications to routing problems and information transmission in networks, where sheaf theory helps understand constraints and cosheaves help understand flows.

4 Methods

The thesis develops several key mathematical tools:

- **Cellular Sheaf Cohomology:** Computation methods for detecting inconsistencies in local data assignments.
- **Cosheaf Homology:** Methods for understanding how local data can be assembled into global structures.

- **Duality Theory:** Deep connections between sheaves and cosheaves through Verdier duality and Poincaré duality.
- **Derived Categories:** Advanced homological algebra tools for studying sheaves and cosheaves systematically.

The approach combines abstract category theory with concrete algorithms, making the theory both mathematically rigorous and computationally practical.

5 Results

The thesis establishes several fundamental results:

- **Decomposition Theorems:** Complete classification of cellular sheaves and cosheaves on graphs and cell complexes.
- **Computational Algorithms:** Efficient methods for computing sheaf cohomology and cosheaf homology.
- **Applications:** Successful applications to sensor networks, network coding, and data analysis problems.
- **Duality Results:** Deep mathematical connections between sheaves and cosheaves that enable transfer of results between domains.

A key insight is that many practical problems in data analysis naturally involve both sheaf-like constraints (consistency requirements) and cosheaf-like structures (assembly processes).

6 Significance

This work provides the mathematical foundation for applying sheaf theory to modern data science problems. It demonstrates that:

- Sheaf theory is not just abstract mathematics but provides practical tools for data analysis.
- The dual concept of cosheaves is equally important for understanding data assembly and flow processes.
- Many problems in sensor networks, persistent homology, and network coding can be unified under this framework.
- The discrete, cellular versions make these ideas computationally accessible.

The thesis opens up new research directions in topological data analysis and provides tools for understanding complex, multi-scale data systems.